Senior Thesis

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```
rhfs <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs <- rhfs |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSPACE, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECCOMMREN, RNTAPPFEEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVLP,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINCTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
   )

# Renaming some factor levels for later convenience
rhfs <- rhfs |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
```

```
      "Assisted living limited interview; self response" = "3",
      "Conversion from partial to full interview" = "5",
      "Interview Compl; field interview" = "201",
      "Assisted living limited interview; field interview" = "203",
      "Sufficient partial; field interview" = "204",
      "Military limited field interview" = "206"
    )
  )

# Rearranging variables greatly improves View(rhfs)
questionnaire_order <- c(
  # 1c-1e and property definition
  "PROPTOWN", "PROPCON", "COMPLEX",
  "NUMBLD_R", "NUMUNITS_R", "NUMCAT_R",

  # 2-4e
  "UNITSTAT", "YRNEWBLG_R", "HOWACQ", "YRPROPACQ_R", "PURPRICE_R",
  "CASHDWN", "CASHDWNAMT", "MRKTVAL_R", "HOWMVAL", "APPRWAIVER",

  # 5-6: occupancy, rents, and rent payment methods
  "BROOMS", "TOTRENTVAC", "TRENOC_R", "TRENVA_R", "TOWNOC_R",
  "TEMPOC_R", "TOTHVA_R", "LOWRENT", "HIRENT", "RENT",
  "RENTNO_R", "RENTLT249_R", "RENT274_R", "RENT299_R", "RENT349_R",
  "RENT399_R", "RENT449_R", "RENT499_R", "RENT549_R", "RENT599_R",
  "RENT649_R", "RENT699_R", "RENT749_R", "RENT799_R", "RENT899_R",
  "RENT999_R", "RENT1099_R", "RENT1249_R", "RENT1499_R", "RENT1999_R",
  "RENT2499_R", "RENT2999_R", "RENT3499_R", "RENT3999_R", "RENT4499_R",
  "RENT4999_R", "RENT5000_R", "BED0_R", "BED1_R", "BED2_R", "BED3_R",
  "PAYONLINE", "PAYTHIRD", "PAYAPP", "PAYCREDIT", "PAYCHK", "PAYCASH", "PAYOTH",

  # 6a-11c
  "LEASLNG", "LNGTRMPRV", "WEEKLNG", "WEEKLNGNUM", "WEEKLNGPCT",
  "RNTAPPFEE_R", "NORNTAPPFEE", "RNTAPPFEEWAIV", "PCTFEEWAIV_R", "SECDEPAMT",
  "CONCSS", "CONCSSNUM", "SENIOR", "RCONTROL", "RCONTROLAHS",
  "OFFSTPARK", "OFFSTPARKAHS", "IELEC", "IGAS", "IWATER", "ISEWER", "ITRASH",
  "ICABLE", "IWIFI", "IPOOL", "IPARKING", "PROPANS",

  # 12a-15e
  "TOTCOLL", "COMMSPACE", "RECCOMMREN", "COMMRENT", "MNGMNT", "HRSMNGMNT",
  "ADVERTISE", "ADWEBSITE", "ADONLINE", "ADOTHONLINE", "ADNEWSPAPER",
  "ADAGENCY", "ADAGENT", "MLS",

  # 16: operating expenses
  "OPPAY", "OPTAX_R", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
  "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX", "OPEX_R",

  # 17-18c: capital expenses and ownership
  "CAPEX_R", "CPHEATAMT", "CPRFAMT", "CPWINAMT", "CPDOORAMT", "CPFLRAMT",
  "CPEXTAMT", "CPPLBAMT", "CPELEAMT", "CPKITAMT", "CPBATAMT", "CPACCAMT",
  "CPOTHAMT", "CPRESAMT_R", "OWNENT", "OWNOTHPROP", "NUMOTHPROP", "NUMOTHUNITS",

  # 19-23: subsidies, tax credits, and tax reductions
  "HUDUNITS", "GOVBELOW", "RSUBSIDY", "ELDERLY", "GOVGRANT", "BENEOTH",
```

```
  "LIHTCUNITS", "STLOWINC", "HISTORIC", "LOWINCTEN", "LOWINAREA", "ECONDEVLP", "OTHRCV",

  # 24-37: debt/mortgage section and HUD voucher history
  "DEBT", "WHYNOMRT", "DEBTNUM", "DEBT1", "YRMORT_R", "MORTAMT_R", "OUTBAL_R",
  "APPVAL_R", "GREENMORT", "MORTGET", "LOCTYP", "REGPAY", "PMTAMT_R", "PMTFREQ",
  "RPIPRIN", "RPIPRPINS", "RPIPMI", "RPIPRPTAX", "ACCEPT"
)

rhfs <- rhfs |>
  relocate(
    CONTROLPUF, OUTCOME, SAMPSTRATA, SURVYR,
    all_of(questionnaire_order)
  ) |>
  # Keep the full-sample and replicate weights together at the far right.
  relocate(
    WEIGHT, starts_with("REPWGT"), .after = last_col()
  )
```

```
# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was
    # done in Census's Methodology)
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )
```

```
## # A tibble: 1 x 3
##   properties properties_se   moe
##        <dbl>         <dbl> <dbl>
## 1     18965.          165.  271.
```

```
# Makes View(rhfs) much more productive.
rhfs <- rhfs |>
  select(-starts_with("REPWGT"), -WEIGHT,
         -SURVYR, -CONTROLPUF, # is always 2024, ID isn't useful in data analysis
         # J flag variables mark values that were manually reviewed; note that
```

```
        # they may become relevant if outliers are under scrutiny.
        -starts_with("J"),
        # A continuous variable coded as a factor without documentation; may
        # be useful for pure prediction
        # -PCTFEEWAIV_R,
        # Poorly documented variable with 99.5% of its values missing
        -WEEKLNGPCT
        )

# NAs in the dataset
missingvalues <- function(df) {
  NAs <- sum(sapply(df, \(x) sum(x == -8))) # Not applicable
  NRs <- sum(sapply(df, \(x) sum(x == -9))) # Not reported

  tibble(
    NAs = NAs, # Not applicable
    NRs = NRs, # Not reported
    perc.missing = (NAs + NRs) / prod(dim(df)) # % of missing values in the df
  )
}
missingvalues(rhfs)
```

```
## # A tibble: 1 x 3
##      NAs    NRs perc.missing
##    <int>  <int>        <dbl>
## 1 170275 144234        0.402
```

```
# Missing values by interview type
rhfs |>
  summarize(
    # pick(everything()) extracts a data frame of all non-grouping columns
    missingvalues(pick(everything())),
    .by = OUTCOME
  )
```

```
##                                           OUTCOME   NAs   NRs perc.missing
## 1                    Interview Compl; field interview 97135 73188    0.3933920
## 2                Sufficient partial; field interview 29948 36570    0.5177304
## 3                     Interview Compl; self response 37769 21028    0.3230888
## 4          Conversion from partial to full interview  3442  7740    0.4538149
## 5                    Military limited field interview   297   839    0.7171717
## 6 Assisted living limited interview; field interview  1286  3761    0.7169034
## 7   Assisted living limited interview; self response   398  1108    0.7130682
```

```
# Handling known Trivially Missing values
# Certain values in the columns below can only exist if REGPAY is 1, so -9 is
# wrongly assigned to them.
trivially.missing.regpay <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    REGPAY == -9 &
      # In a given observation, if any of these columns contain a value other than
```

```
      # -8 or -9 return True.
      if_any(c(PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, REGPAY, PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX) |>
  pull(id)

rhfs$REGPAY[trivially.missing.regpay] <- 1


trivially.missing.ownothprop <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OWNOTHPROP == -9 &
      if_any(c(NUMOTHPROP, NUMOTHUNITS), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, NUMOTHPROP, NUMOTHUNITS) |>
  pull(id)

rhfs$OWNOTHPROP[trivially.missing.ownothprop] <- 1


trivially.missing.rntappfeewaiv <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    RNTAPPFEEWAIV == -9 &
      if_any(c(PCTFEEWAIV_R), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, PCTFEEWAIV_R) |>
  pull(id)

rhfs$RNTAPPFEEWAIV[trivially.missing.rntappfeewaiv] <- 1


# Total rental receipts as a percent of property value (TRENTRCTPV_R) is
# calculated with (TOTCOLL + COMMRENT) / MRKTVAL. If COMMSPACE = 2, then
# COMMRENT = 0, so TRENTRCTPV_R = TOTCOLL / MRKTVAL as derived below.
trivially.missing.trentrctpv_r <- rhfs |>
  mutate(id = row_number()) |>
  select(id, TRENTRCTPV_R, COMMSPACE, COMMRENT, RECCOMMREN, TOTCOLL, MRKTVAL_R) |>
  filter(TRENTRCTPV_R == -8 & COMMSPACE == 2 &
      if_all(c(TOTCOLL, MRKTVAL_R), \(x) !x %in% c(-8, -9))
    ) |>
  pull(id)

rhfs$TRENTRCTPV_R[trivially.missing.trentrctpv_r] <-
  rhfs$TOTCOLL[trivially.missing.trentrctpv_r] /
  rhfs$MRKTVAL_R[trivially.missing.trentrctpv_r]


# Ensuring that when CASHDWN is 2, CASHDWNAMT is indeed always -9
rhfs |>
  mutate(id = row_number()) |>
  select(id, CASHDWN, CASHDWNAMT) |>
```

```
  filter(CASHDWN == 2 & CASHDWNAMT != -9)
```

```
## [1] id         CASHDWN    CASHDWNAMT
## <0 rows> (or 0-length row.names)
```

```
# If a CASHDWN payment wasn't made, then the CASHDWNAMT should be 0
rhfs$CASHDWNAMT <- ifelse(rhfs$CASHDWN == 2, 0, rhfs$CASHDWNAMT)


# Same idea as with CASHDWN/CASHDWNAMT
rhfs |>
  mutate(id = row_number()) |>
  select(id, WEEKLNG, WEEKLNGNUM) |>
  filter(WEEKLNG == 2 & WEEKLNGNUM != -9)
```

```
##     id WEEKLNG WEEKLNGNUM
## 1 2173       2          0
## 2 3507       2          0
## 3 4197       2          0
```

```
# If no units are offered for daily/weekly rental, then WEEKLNGNUM (the count
# of those that are offered) should be 0
rhfs$WEEKLNGNUM <- ifelse(rhfs$WEEKLNG == 2, 0, rhfs$WEEKLNGNUM)


# Viewing targets for imputation
rhfs |>
  mutate(id = row_number()) |>
  select(COMMSPACE, COMMRENT, RECCOMMREN) |>
  filter(COMMSPACE == 2)
```

```
##      COMMSPACE COMMRENT RECCOMMREN
## 1            2       -9         -9
## 2            2       -9         -9
## 3            2       -9         -9
## 4            2       -9         -9
## 5            2       -9         -9
## 6            2       -9         -9
## 7            2       -9         -9
## 8            2       -9         -9
## 9            2       -9         -9
## 10           2       -9         -9
## 11           2       -9         -9
## 12           2       -9         -9
## 13           2       -9         -9
## 14           2       -9         -9
## 15           2       -9         -9
## 16           2       -9         -9
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## 18           2       -9         -9
## 19           2       -9         -9
## 20           2       -9         -9
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## 21          2       -9        -9
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## 3459        2       -8        -9
## 3460        2       -8        -9
## 3461        2       -8        -9
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## 3465        2       -8        -9
## 3466        2       -8        -9
## 3467        2       -8        -9
## 3468        2       -8        -9
## 3469        2       -8        -9
## 3470        2       -8        -9
## 3471        2       -8        -9
## 3472        2       -8        -9
## 3473        2       -8        -9
## 3474        2       -8        -9
## 3475        2       -8        -9
## 3476        2       -8        -9
```

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## 3477        2       -8        -9
## 3478        2       -8        -9
## 3479        2       -8        -9
## 3480        2       -8        -9
## 3481        2       -8        -9
## 3482        2       -8        -9
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## 3486        2       -8        -9
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## 3488        2       -8        -9
## 3489        2       -8        -9
## 3490        2       -8        -9
## 3491        2       -8        -9
## 3492        2       -8        -9
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## 3495        2       -8        -9
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## 3501        2       -8        -9
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## 3514        2       -8        -9
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## 3516        2       -8        -9
## 3517        2       -8        -9
## 3518        2       -8        -9
## 3519        2       -8        -9
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## 3523        2       -8        -9
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## 3526        2       -8        -9
## 3527        2       -8        -9
## 3528        2       -8        -9
## 3529        2       -8        -9
## 3530        2       -8        -9
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## 3532        2       -8        -9
## 3533        2       -8        -9
## 3534        2       -8        -9
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## 3536        2       -8        -9
## 3537        2       -8        -9
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## 3542        2       -8        -9
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## 3551        2       -8        -9
## 3552        2       -8        -8
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## 3554        2       -8        -9
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## 3559        2       -8        -9
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## 3561        2       -8        -9
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## 3563        2       -8        -9
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## 3565        2       -8        -9
## 3566        2       -8        -9
## 3567        2       -8        -9
## 3568        2       -8        -9
## 3569        2       -8        -9
## 3570        2       -8        -9
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## 3573        2       -8        -9
## 3574        2       -8        -9
## 3575        2       -8        -9
## 3576        2       -8        -9
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## 3579        2       -8        -9
## 3580        2       -8        -9
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## 3582        2       -8        -9
## 3583        2       -8        -9
## 3584        2       -8        -9
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## 3586        2       -8        -9
## 3587        2       -8        -9
## 3588        2       -8        -9
## 3589        2       -8        -9
## 3590        2       -8        -9
## 3591        2       -8        -9
## 3592        2       -8        -9
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## 3599        2       -8        -9
## 3600        2       -8        -9
## 3601        2       -8        -9
## 3602        2       -8        -9
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## 3638        2       -8        -9
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## 3639        2       -8        -9
## 3640        2       -8        -9
## 3641        2       -8        -9
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## 3649        2       -8        -9
## 3650        2       -8        -9
## 3651        2       -8        -9
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## 3657        2       -8        -9
## 3658        2       -8        -9
## 3659        2       -8        -9
## 3660        2       -8        -9
## 3661        2       -8        -9
## 3662        2       -8        -9
## 3663        2       -8        -9
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## 3665        2       -8        -9
## 3666        2       -8        -9
## 3667        2       -8        -9
## 3668        2       -8        -9
## 3669        2       -8        -9
## 3670        2       -8        -9
## 3671        2       -8        -9
## 3672        2       -8        -9
## 3673        2       -8        -9
## 3674        2       -8        -9
## 3675        2       -8        -9
## 3676        2       -8        -9
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## 3679        2       -8        -9
## 3680        2       -8        -9
## 3681        2       -8        -9
## 3682        2       -8        -9
## 3683        2       -8        -9
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## 3685        2       -8        -9
## 3686        2       -8        -9
## 3687        2       -8        -9
## 3688        2       -8        -9
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## 3692        2       -8        -9
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## 3693        2       -8        -9
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## 3698        2       -8        -9
## 3699        2       -8        -9
## 3700        2       -8        -9
## 3701        2       -8        -9
## 3702        2       -8        -9
## 3703        2       -8        -9
## 3704        2       -8        -9
## 3705        2       -8        -9
## 3706        2       -8        -9
## 3707        2       -8        -9
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## 3709        2       -8        -9
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## 3731        2       -8        -9
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## 3743        2       -8        -9
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## 3746        2       -8        -9
```

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## 3747        2       -8        -9
## 3748        2       -8        -9
## 3749        2       -8        -9
## 3750        2       -8        -9
## 3751        2       -8        -9
## 3752        2       -8        -9
## 3753        2       -8        -9
## 3754        2       -8        -9
## 3755        2       -8        -9
## 3756        2       -8        -9
## 3757        2       -8        -9
## 3758        2       -8        -9
## 3759        2       -8        -9
## 3760        2       -8        -9
## 3761        2       -8        -9
## 3762        2       -8        -9
## 3763        2       -8        -9
## 3764        2       -8        -9
## 3765        2       -8        -9
## 3766        2       -8        -9
## 3767        2       -8        -9
## 3768        2       -8        -9
## 3769        2       -8        -9
## 3770        2       -8        -9
## 3771        2       -8        -9
## 3772        2       -8        -9
## 3773        2       -8        -9
## 3774        2       -8        -9
## 3775        2       -8        -9
## 3776        2       -8        -9
## 3777        2       -8        -9
## 3778        2       -8        -9
## 3779        2       -8        -9
## 3780        2       -8        -9
## 3781        2       -8        -9
## 3782        2       -8        -9
## 3783        2       -8        -9
## 3784        2       -8        -9
## 3785        2       -8        -9
## 3786        2       -8        -8
## 3787        2       -8        -9
## 3788        2       -8        -9
## 3789        2       -8        -9
## 3790        2       -8        -9
## 3791        2       -8        -9
## 3792        2       -8        -9
## 3793        2       -8        -9
## 3794        2       -8        -9
## 3795        2       -8        -9
## 3796        2       -8        -9
## 3797        2       -8        -9
## 3798        2       -8        -8
## 3799        2       -8        -9
## 3800        2       -8        -9
```

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## 3801         2       -8         -9
## 3802         2       -8         -9
## 3803         2       -8         -9
## 3804         2       -8         -9
## 3805         2       -8         -9
## 3806         2       -8         -9
## 3807         2       -8         -9
## 3808         2       -8         -9
## 3809         2       -8         -9
## 3810         2       -8         -9
## 3811         2       -8         -9
## 3812         2       -8         -9
## 3813         2       -8         -9
## 3814         2       -8         -9
## 3815         2       -8         -9
## 3816         2       -8         -9
## 3817         2       -8         -9
```

```
# If there is no commercial space on this property, then the property did not
# receive any rental income from a commercial use of the property.
rhfs$RECCOMMREN[rhfs$COMMSPACE == 2] <- "2"
rhfs$COMMRENT <- ifelse(rhfs$COMMSPACE == 2, 0, rhfs$COMMRENT)


rhfs |>
  mutate(id = row_number()) |>
  select(RECCOMMREN, COMMRENT) |>
  filter(RECCOMMREN == 2) |>
  summarize(
    n(),
    .by = COMMRENT
  )
```

```
##   COMMRENT  n()
## 1        0 3874
## 2       -9    9
```

```
# If commercial rent was not received, then commercial rent should be 0
rhfs$COMMRENT <- ifelse(rhfs$RECCOMMREN == 2, 0, rhfs$COMMRENT)


# More informative than "-8"
rhfs <- rhfs |>
  mutate(
    UNITSTAT = fct_recode(
      UNITSTAT,
      "Not a single unit" = "-8"
    )
  )



missing_by_var <- data.frame(
```

```
  variable = names(rhfs),
  percent_missing = sapply(
    rhfs,
    \(x) mean(x %in% c(-9))
  )
) |>
  arrange(desc(percent_missing))



# Evidence for MAR (missing at random) in MRKTVAL_R
df |>
  mutate(
    mrktNA = MRKTVAL_R == -9
  ) |>
  summarize(
    survey_mean(mrktNA),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 3
##   NUMCAT_R      coef  `_se`
##   <fct>        <dbl>  <dbl>
## 1 1 Unit      0.0923 0.0263
## 2 2-4 Units   0.0716 0.0224
## 3 5-24 Units  0.154  0.0224
## 4 25-49 Units 0.335  0.0547
## 5 50+ Units   0.418  0.0163
```

```
df |>
  filter(MRKTVAL_R != -9) |>
  summarize(
    mean_mrktval = survey_mean(MRKTVAL_R),
    median_mrktval = survey_median(MRKTVAL_R),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 5
##   NUMCAT_R    mean_mrktval mean_mrktval_se median_mrktval median_mrktval_se
##   <fct>              <dbl>           <dbl>          <dbl>              <dbl>
## 1 1 Unit           397050.          42218.         250000             33154.
## 2 2-4 Units        569582.          33304.         425000             50491.
## 3 5-24 Units      2178309.         118540.        2000000            469164.
## 4 25-49 Units     5000928.         333704.        4000000            224403.
## 5 50+ Units      33895580.        2865340.       20000000           4987831.
```

```
write.csv(rhfs, "RHFS-2024-Cleaned.csv", row.names = FALSE)
```